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Advanced Financial Portfolio Management System Using Object-Oriented Programming Techniques in C++

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA0109-Object Oriented Programming with C++

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

AI & ML

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Advanced Financial Portfolio Management System Using Object-Oriented Programming Techniques in C++**” has been carried out by Sravanthi.D(192425315) under the supervision of Dr.I.Sudha and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech AI&ML** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **D.Sravanthi (192425315)** of the AI&ML, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Advanced Financial Portfolio Management System Using Object-Oriented Programming Techniques in C++**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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(Mandatory for all team projects)

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ABSTRACT

Managing personal investments effectively can be challenging when investment information is maintained across different platforms, spreadsheets, or applications. Investors may find it difficult to track their investments, calculate portfolio performance, understand risk levels, and obtain suitable investment insights from a centralized system. The absence of an integrated portfolio management solution can lead to poor visibility of financial performance and difficulty in making informed investment decisions.

FinSight AI – Advanced Financial Portfolio Management and Intelligent Investment Recommendation Web Application is proposed to address these challenges through a centralized web-based financial management system. The application is designed with three major modules: **Portfolio Management, Financial Analysis, and Risk & Recommendation**. The Portfolio Management module enables users to maintain their profile and add, view, update, delete, search, and filter investment records across different asset categories. The Financial Analysis module processes investment data to calculate total investment, current portfolio value, profit or loss, and Return on Investment (ROI). The Risk & Recommendation module applies predefined prototype rules to classify investment risk and provide basic investment recommendations.

The methodology involves gathering and preparing an investment dataset, designing a structured portfolio management interface, implementing the application modules, applying financial calculations and rule-based analysis, and testing the system using sample investment records. Graphs and charts are incorporated into the dashboard to provide a clear visual representation of portfolio performance and asset allocation. Testing with prototype data demonstrated that the application can successfully organize investment records, perform portfolio calculations, display financial results, and provide basic risk-based recommendations.

The project concludes that FinSight AI provides a practical and user-friendly foundation for centralized investment portfolio management and financial analysis. It improves the accessibility and understanding of investment information through automated calculations and visual insights. The system can be further enhanced by integrating real-time market data,

advanced machine learning models, personalized recommendations, and secure financial-data services.

Keywords: Financial Portfolio Management, Investment Analysis, Risk Assessment, ROI, Intelligent Recommendation, Web Application

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
ROI	Return on Investment	%
ETF	Exchange-Traded Fund	—
AI	Artificial Intelligence	—
ML	Machine Learning	—
CRUD	Create, Read, Update, Delete	—
C++	Object-Oriented Programming Language	—
UI	User Interface	—
API	Application Programming Interface	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Investment management involves maintaining and monitoring information related to different financial assets such as stocks, mutual funds, exchange-traded funds (ETFs), bonds, fixed deposits, and cryptocurrencies. When an investor holds multiple investments, it becomes important to maintain accurate records of the quantity, purchase price, current value, profit or loss, and overall portfolio performance. Managing these details manually can make it difficult to obtain a clear and consolidated view of the portfolio.

FinSight AI – Advanced Financial Portfolio Management and Intelligent Investment Recommendation Web Application is proposed as a centralized web-based solution for managing and analysing investment portfolios. The system integrates portfolio management, financial analysis, risk classification, recommendations, and graphical visualization into a single application.

The application consists of three major modules. **Module 1 – Portfolio Management** enables users to add, view, update, delete, search, and filter investment records. **Module 2 – Financial Analysis** processes investment information to calculate total investment, current portfolio value, profit or loss, and Return on Investment (ROI). **Module 3 – Risk and Recommendation** provides prototype-based risk classification and basic investment recommendations based on the available portfolio information.

The system uses structured investment data containing details such as investment name, investment type, quantity or units, purchase price, and current price or value. Automated calculations reduce the possibility of manual calculation errors, while graphs and charts provide a visual representation of portfolio performance and asset distribution.

FinSight AI is developed as an academic web application prototype demonstrating the integration of software development, structured data management, financial computation, risk analysis, recommendation logic, and data visualization in a unified system.

1.2 Need for the Project

Investment portfolios may contain several assets with different quantities, prices, values, and risk characteristics. Maintaining these details manually through spreadsheets or separate applications can be time-consuming and may result in calculation errors, outdated records, or difficulty in understanding the overall portfolio performance. Therefore, a centralized system is required to organize investment information and simplify portfolio analysis.

The problem primarily affects individual investors and users who manage multiple investments and require a convenient way to monitor their portfolio. Beginner investors can also benefit from a simplified application that presents investment information, financial calculations, risk categories, and portfolio insights in an understandable format.

Existing approaches such as spreadsheets and manual calculations generally require users to enter and update information themselves. They may also require separate calculations for investment value, profit or loss, and ROI. Using different platforms for portfolio records, analysis, and visualization can make the overall process less efficient.

From an engineering perspective, the project involves several interacting components, including data management, user interface design, financial calculations, risk classification, recommendation logic, and graphical visualization. The system must maintain accurate calculations and consistent investment records while providing a simple and responsive user experience.

The engineering significance of the project lies in developing an integrated web-based solution that demonstrates how structured financial data can be processed and converted into useful portfolio information through automated computation and rule-based analysis.

1.3 Problem Statement

Managing a diversified investment portfolio requires the effective handling of investment records, asset categories, quantities, purchase prices, current values, financial calculations, risk levels, recommendations, and portfolio visualizations. Manual and disconnected approaches can make it difficult to maintain accurate records and obtain a consolidated understanding of portfolio performance.

The engineering problem is to **design and develop a centralized web-based financial portfolio management system capable of managing multiple investment records, automatically calculating portfolio performance, classifying investment risk, generating basic recommendations, and presenting financial information through graphical visualizations while maintaining data accuracy, consistency, usability, security, and scalability within realistic academic project constraints.**

The system must handle different investment categories and their associated information while balancing functionality and simplicity. It should provide accurate calculations using the available data and generate basic risk and recommendation outputs without representing prototype recommendations as guaranteed financial advice.

1.4 Project Aim

The aim of this project is to **design and develop FinSight AI, a centralized web-based financial portfolio management and intelligent investment recommendation application that enables users to manage investments, analyse portfolio performance, assess investment risk, and obtain basic portfolio insights through automated calculations and visualizations.**

1.5 Project Objectives

1. **To design** a user-friendly web-based interface for centralized financial portfolio management.
2. **To develop** a Portfolio Management module for adding, viewing, updating, deleting, searching, and filtering investment records.
3. **To gather and prepare** a structured investment dataset containing the required information for portfolio management and analysis.
4. **To model** investment information using relevant attributes such as investment type, quantity, purchase price, and current price or value.
5. **To implement** automated calculations for total investment, current portfolio value, profit or loss, and Return on Investment (ROI).
6. **To develop** a prototype-based risk classification mechanism for categorizing different investment types according to their risk levels.
7. **To implement** a basic recommendation component that generates portfolio insights based on investment and risk information.
8. **To develop** graphical visualizations for displaying portfolio performance and asset allocation.
9. **To test** the application using sample investment records and verify the correctness of portfolio management, financial calculations, risk classification, recommendations, and visual outputs.
10. **To evaluate** the developed system against the identified requirements and analyse its strengths, limitations, and potential future enhancements.

1.6 Scope

1.6.1 Included

The scope of FinSight AI includes the development of a web-based investment portfolio management and analysis application. The system includes user profile management and

portfolio record management for different investment categories such as stocks, mutual funds, ETFs, bonds, fixed deposits, cryptocurrencies, and other investments.

The application includes functionality for adding, viewing, updating, deleting, searching, and filtering investment records. It also performs automated financial calculations including total investment, current value, profit or loss, and ROI. Dashboard visualizations are provided to represent portfolio performance and asset allocation.

The system also includes prototype-based risk classification and basic recommendation features using the available investment information.

1.6.2 Excluded

The project does not include actual buying or selling of financial assets, money transfers, brokerage services, or execution of investment transactions. It does not guarantee investment returns or provide professional financial advice. Real-time market data integration and advanced predictive investment models are considered future enhancements unless specifically implemented in the current prototype.

1.6.3 Assumptions

The system assumes that the investment information entered by the user is accurate and sufficiently complete for portfolio calculations. The prototype also assumes that quantity, purchase price, and current price or value are available for calculating investment performance. The risk classification and recommendations are assumed to be educational prototype outputs based on predefined rules and should not be considered professional financial recommendations.

1.6.4 Boundary Conditions

The application operates within the available investment data provided by the user or prepared dataset. Financial calculations depend on the accuracy of the entered values. The system is designed primarily for portfolio management, analysis, visualization, and prototype recommendation rather than real-world financial transaction execution.

1.7 Expected Engineering Outcome

The expected engineering outcome of the project is a **functional web-based prototype named FinSight AI** that integrates portfolio management, financial analysis, risk classification, recommendation logic, and visualization.

The expected outcomes are:

- **System:** A centralized web application for managing and analysing investment portfolios.

- **Product:** A user-friendly portfolio management and financial analysis application.
- **Process:** A structured process for gathering investment data, maintaining portfolio records, performing financial calculations, analysing risk, and presenting results.
- **Prototype:** A working academic prototype demonstrating the integration of portfolio management and intelligent financial insights.
- **Algorithm/Logic:** Automated financial calculation formulas and predefined rule-based risk and recommendation logic.
- **Model:** A structured representation of investment records and portfolio performance.
- **Infrastructure Solution:** A web-based architecture capable of organizing investment data and presenting financial information through an integrated dashboard.

The final system is expected to demonstrate improved organization of investment records, automated portfolio calculations, simplified financial analysis, and clear visualization of portfolio information. It also provides a foundation for future enhancements such as real-time market-data integration, advanced machine-learning-based recommendations, and more sophisticated risk-analysis techniques.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review for the FinSight AI project focused on existing approaches for financial portfolio management, investment analysis, risk assessment, recommendation systems, and web-based financial applications. Relevant research papers, existing financial management systems, commercial investment platforms, data-analysis techniques, and technologies used for portfolio analysis were considered.

The review was carried out by examining the commonly used approaches for recording investment information, calculating portfolio performance, analysing investment risk, and presenting financial information through dashboards and visualizations. Particular attention was given to the advantages and limitations of centralized portfolio management systems and automated financial-analysis methods.

The reviewed approaches were then compared based on portfolio management functionality, financial analysis, risk assessment, recommendation capability, visualization, usability, and suitability for an academic web-based application. The findings were used to identify the engineering gap and define the requirements of the proposed FinSight AI system.

2.2 Review of Existing Work

2.2.1 Portfolio Management Systems

Existing portfolio management systems provide facilities for maintaining investment records and monitoring portfolio performance. These systems generally allow users to record assets, quantities, purchase prices, current values, and other investment-related information. They reduce the need for maintaining separate manual records and provide a centralized view of investments.

However, the functionality and complexity of existing systems vary considerably. Professional platforms may provide extensive financial information and advanced services, but they can include features that are unnecessary for a simple academic portfolio-management application. The proposed FinSight AI system focuses on providing essential portfolio management features through a simplified and understandable interface.

2.2.2 Financial Analysis Methods

Financial portfolio analysis commonly involves calculating the total amount invested, current portfolio value, profit or loss, and percentage return. These calculations provide users with a basic understanding of portfolio performance.

For FinSight AI, these calculations are automated using structured investment data. The investment amount is calculated from the quantity and purchase price, while the current portfolio value is determined using the quantity and current price or value. Profit or loss and ROI are then calculated from these values.

This approach reduces repetitive manual calculations and provides consistent results across investment records.

2.2.3 Risk Assessment Approaches

Investment risk can vary according to the type of financial asset. Existing financial systems may use historical performance, market volatility, financial indicators, investor profiles, or statistical models to estimate risk.

For the academic prototype, FinSight AI uses a simplified rule-based approach. Different investment types are mapped to prototype risk categories such as Low, Medium, High, and Very High. This approach is easier to implement and understand while demonstrating the concept of automated risk classification.

The limitation of this approach is that investment risk in real financial markets depends on several factors and cannot be completely represented by asset type alone.

2.2.4 Recommendation Systems

Recommendation systems are widely used to provide personalized suggestions based on available data. In financial applications, recommendations can potentially consider portfolio composition, investment performance, risk tolerance, historical trends, and other financial factors.

FinSight AI incorporates a basic recommendation component based on portfolio information and prototype risk classification. The purpose is to demonstrate how investment information can be transformed into simple portfolio insights. Advanced machine learning, predictive analytics, and personalized financial advisory models are considered possible future enhancements.

2.2.5 Financial Dashboards and Visualization

Financial dashboards provide graphical representations of important information such as asset allocation, portfolio value, profit or loss, and investment distribution. Visualization makes numerical information easier to understand and enables users to identify portfolio patterns quickly.

FinSight AI incorporates graphical representations of portfolio information through dashboard charts. These visualizations complement the numerical calculations and provide users with a consolidated view of their investments.

2.2.6 Web-Based Technologies

Web applications are suitable for financial portfolio management because they provide centralized access through a browser and allow different application components to be integrated into a single interface. A web-based architecture also provides flexibility for future improvements such as database integration, authentication, real-time market data, and advanced analytics.

The proposed application uses web technologies to provide an interactive interface for portfolio management, financial analysis, risk assessment, recommendations, and visualization.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Record Keeping	Simple and inexpensive	Time-consuming and prone to errors	Provides the basic problem addressed by FinSight AI
Spreadsheet-Based Portfolio Management	Flexible and easy to customize	Requires manual updates and calculations	FinSight AI automates portfolio operations and calculations

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Basic Portfolio Tracking Applications	Centralized investment records	Limited analysis and recommendation features	Provides reference for portfolio-management functionality
Commercial Investment Platforms	Advanced financial information and features	Can be complex for basic users	Provides reference for required portfolio-management features
Financial Analysis Tools	Automated financial calculations	May focus only on analysis	Financial analysis is integrated with portfolio management
Rule-Based Risk Assessment	Simple, transparent, and easy to implement	Cannot represent all real-world market factors	Used for prototype risk classification
Advanced Financial Recommendation Systems	Data-driven and personalized insights	Require complex models and larger datasets	Provides direction for future recommendation features
Financial Dashboards	Easy graphical representation of information	Visualization alone does not provide complete portfolio management	FinSight AI combines visualization with management and analysis

2.4 Research / Engineering Gap

The review indicates that existing approaches provide individual solutions for portfolio tracking, financial calculation, risk analysis, recommendation, or visualization. However, there is a need for a simplified integrated system that combines these functions within a single academic web application.

Manual and spreadsheet-based approaches require considerable user involvement for maintaining records and performing calculations. Commercial platforms may provide more advanced functionality but can be complex for users who only require fundamental portfolio-management and analysis features. Basic tracking systems may also focus primarily on storing investment information without combining portfolio management, automated financial calculations, risk classification, recommendations, and visualization.

Another important gap is the balance between **functionality and simplicity**. A system intended for academic use should demonstrate meaningful engineering concepts without introducing unnecessary complexity. At the same time, financial recommendations must be presented carefully because simplified rules cannot represent the complete complexity of real-world financial markets.

Therefore, the identified engineering gap is the absence of a simple, centralized, and integrated prototype that combines investment record management, automated portfolio analysis, basic risk classification, recommendation logic, and graphical visualization in a single web-based application.

2.5 Proposed Contribution

FinSight AI addresses the identified gap by integrating portfolio management, financial analysis, risk classification, recommendation logic, and visualization into a single web-based application.

The proposed system contributes the following:

1. **Centralized Portfolio Management:** Provides facilities to add, view, update, delete, search, and filter investment records from one application.
2. **Structured Investment Dataset:** Provides organized investment information containing relevant attributes required for portfolio management and financial analysis.
3. **Automated Financial Analysis:** Calculates investment amount, current value, profit or loss, and ROI automatically from the available investment data.
4. **Prototype Risk Classification:** Applies predefined rules to classify investments into different risk categories according to investment type.
5. **Basic Intelligent Recommendations:** Generates simple portfolio insights based on available investment and risk information.

6. **Graphical Visualization:** Presents portfolio performance and asset allocation using charts and dashboard visualizations.
7. **Integrated Web Application:** Combines the major functions into a single user-friendly web interface rather than requiring separate tools for portfolio records, calculations, and visualization.

The proposed contribution is therefore not intended to replace professional financial platforms. Instead, it provides a practical academic prototype demonstrating how financial data management, automated computation, rule-based analysis, recommendation logic, and visualization can be integrated into a unified engineering solution.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
End User / Investor	Add and maintain investment records in a centralized application.
End User / Investor	View portfolio details and current investment information.
End User / Investor	Update or delete existing investment records.
End User / Investor	Search and filter investments based on available attributes.
End User / Investor	View total investment, current value, profit/loss, and ROI.
End User / Investor	Understand portfolio distribution through graphical visualizations.
End User / Investor	View basic risk classification and investment recommendations.
Project Team	Maintain accurate and structured investment data for application processing and testing.

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Add investment records	Functional	New investment record stored successfully.
View investment records	Functional	All available records displayed correctly.
Update investment records	Functional	Selected record updated without affecting other records.
Delete investment records	Functional	Selected record removed successfully.
Search and filter investments	Functional	Matching records displayed based on selected criteria.
Calculate investment value	Functional	Quantity $\times$ Purchase Price calculated correctly.
Calculate current portfolio value	Functional	Quantity $\times$ Current Price calculated correctly.
Calculate profit/loss and ROI	Functional	Results calculated accurately from stored values.
Risk classification	Functional	Investment assigned to predefined risk category.
Recommendation generation	Functional	Basic recommendation generated from portfolio/risk information.
Dashboard visualization	Functional	Portfolio information represented through charts/graphs.

Requirement	Type (Functional/Non-Functional)	Measurable Target
Usability	Non-Functional	Main functions accessible through a clear web interface.
Performance	Non-Functional	Normal user operations respond without noticeable delay.
Data consistency	Non-Functional	Stored and displayed investment values remain consistent.
Security	Non-Functional	User input is validated and sensitive information is protected.
Maintainability	Non-Functional	Modules and application components remain organized for future modification.

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010 – Software Product Quality Model	Consider software quality characteristics such as usability, reliability, performance, security, and maintainability.	These quality factors were considered while designing the web application and its modules.
OWASP Application Security Verification Standard (ASVS)	Follow secure web-application development practices.	Input validation, controlled data handling, and protection of application information are considered during development.
WCAG 2.2	Improve accessibility and usability of web content.	Clear labels, readable content, logical navigation, and

Standard / Code	Requirement	How Applied in This Project
		understandable interface elements are used where applicable.
Data Privacy Principles	Financial information should be handled responsibly and only for the intended application purpose.	The prototype avoids unnecessary collection of personal financial information and uses investment data only for portfolio analysis.

Project Constraints

The project is developed within the constraints of an academic prototype, including limited development time, available computing resources, limited real-world financial datasets, and the need to maintain a simple user interface. The prototype does not execute financial transactions and depends on the accuracy of investment information entered or supplied for analysis.

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Investment Categories	Support stocks, mutual funds, ETFs, bonds, fixed deposits, cryptocurrency, and other investments	Categories	High
Investment Records	Add, view, update, delete records	Records	High
Search and Filter	Search/filter stored investments	Operation	High
Investment Calculation	$\text{Quantity} \times \text{Purchase Price}$	Currency	High
Current Value Calculation	$\text{Quantity} \times \text{Current Price}$	Currency	High

Parameter	Target/Requirement	Unit	Priority
Profit/Loss Calculation	Current Value – Investment Amount	Currency	High
ROI Calculation	$(\text{Profit/Loss} \div \text{Investment Amount}) \times 100$	Percentage	High
Risk Classification	Low, Medium, High, Very High	Category	Medium
Recommendation	Basic risk-based portfolio insight	Output	Medium
Visualization	Portfolio performance and asset allocation charts	Charts	Medium
Web Interface	Clear and user-friendly dashboard	Interface	High
Data Accuracy	Correct processing of entered investment values	Percentage	High
Maintainability	Modular application structure	Quality	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Spreadsheet-Based Portfolio Management

A spreadsheet-based solution can store investment records and perform basic financial calculations using formulas. It is inexpensive and flexible but requires considerable manual data entry and does not provide an integrated web-based user experience.

Alternative 2: Web-Based Portfolio Management Application

A web-based application provides a centralized interface for investment management, automated calculations, dashboard visualization, and recommendation features. It provides better integration and usability while requiring more development effort than a spreadsheet.

Alternative 3: Mobile-Based Portfolio Management Application

A mobile application can provide convenient access to investment information through smartphones. However, it requires platform-specific development and may involve additional development and maintenance effort compared with the selected web-based approach.

Evaluation Matrix

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	30%	3	5	4
Cost	20%	5	4	3
Safety	15%	3	4	4
Sustainability	15%	4	5	3
Reliability	20%	3	5	4
Weighted Score	100%	3.60	4.65	3.65

Scale: 1 = Poor, 2 = Fair, 3 = Moderate, 4 = Good, 5 = Excellent.

3.5 Selected Design & Detailed Design

Based on the evaluation matrix in Section 3.4, **Alternative 2 – Web-Based Portfolio Management Application** was selected because it achieved the highest weighted score. The web-based approach provides better integration of portfolio management, financial analysis, visualization, and recommendation features while maintaining reasonable development cost and scalability.

Architecture / System Flow

User → Web Interface → Portfolio Management → Investment Data → Financial Analysis → Risk & Recommendation → Dashboard Visualization

Key Engineering Calculations

Investment Amount

Investment Amount = Quantity × Purchase Price

Current Portfolio Value

Current Value = Quantity × Current Price

Profit/Loss

Profit/Loss = Current Value – Investment Amount

Return on Investment

$ROI = (\text{Profit/Loss} \div \text{Investment Amount}) \times 100$

For example, if 10 units are purchased at ₹2,500 per unit and the current price is ₹2,750:

Investment Amount = $10 \times ₹2,500 = ₹25,000$

Current Value = $10 \times ₹2,750 = ₹27,500$

Profit = $₹27,500 - ₹25,000 = ₹2,500$

$ROI = (₹2,500 \div ₹25,000) \times 100 = 10\%$

These values are prototype/demo calculations used to verify the application logic.

Systems Approach

The system is divided into interconnected modules so that each component performs a specific function while contributing to the overall portfolio-management process. The Portfolio Management module collects and maintains investment records, which are then used by the Financial Analysis module for automated calculations. The resulting portfolio information is passed to the Risk and Recommendation module for prototype-based analysis. Finally, the processed information is presented through the dashboard using numerical summaries and graphical visualizations.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Web application instead of spreadsheet	Spreadsheet	Automated and integrated user experience	Requires more development effort	Web application selected because it integrates all major modules.
Modular three-module architecture	Single combined module	Better organization and maintainability	Requires interface between modules	Three modules selected for clear

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
				separation of responsibilities.
Rule-based risk classification	Advanced ML risk model	Simple, transparent, and suitable for prototype	Less accurate for real-world market conditions	Rule-based approach selected due to academic scope and available data.
Dashboard visualization	Text-only output	Easier interpretation of portfolio information	Requires additional interface development	Visualization selected because financial trends and allocation are easier to understand graphically.
Structured dataset	Unstructured records	Consistent calculations and processing	Requires data preparation	Structured dataset selected to improve data consistency and analysis.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Minimize unnecessary physical documentation and support digital portfolio management.	Web-based processing reduces dependence on paper-based investment records.	Digital implementation was selected over a manual record-based approach.
Ethics	Recommendations should not be presented	Prototype recommendations are generated using	The system clearly treats recommendations as

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
	as guaranteed financial advice.	predefined rules rather than professional financial analysis.	basic educational insights.
Health & Safety	Avoid design features that encourage unsafe or compulsive financial activity.	The application does not execute trades or financial transactions.	The system is restricted to portfolio management and analysis.
Security	Protect investment information and validate user-provided data.	Web application input and data-handling requirements were considered during development.	Input validation and controlled data handling are included in the design.
Privacy	Avoid unnecessary collection and exposure of personal financial information.	Only information required for portfolio management is considered.	Data collection is limited to project requirements.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect investment data entered by user	Medium	High	High	Validate required fields and numerical values before processing.	Medium

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect financial calculation	Low	High	Medium	Verify formulas using sample calculations and test cases.	Low
Loss or corruption of portfolio data	Medium	High	High	Use structured data storage and maintain backup copies where applicable.	Medium
Unauthorized access to portfolio information	Low	High	Medium	Apply authentication and controlled access where implemented.	Low
Incorrect risk classification	Medium	Medium	Medium	Clearly define and test prototype risk rules.	Low
Misinterpretation of recommendations	Medium	High	High	Clearly state that recommendations are prototype/educational insights and not guaranteed financial advice.	Medium
Outdated market information	Medium	Medium	Medium	Treat current-value information as user-provided/prototype data unless real-time integration is implemented.	Medium
Web application errors	Medium	Medium	Medium	Perform functional and interface testing before final deployment.	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

FinSight AI was developed using an iterative web application development approach. The system was divided into three major modules: Portfolio Management, Financial Analysis, and Risk & Recommendation. The development process involved requirement analysis, interface design, dataset preparation, module implementation, integration, functional testing, and result verification. Sample investment records were used to verify portfolio operations, financial calculations, risk classification, recommendations, and dashboard visualizations.

Resource	Purpose
Investment Dataset	Portfolio management, calculations, testing, and analysis
Computer/Laptop	Application development and testing
Visual Studio Code	Source-code development and project management
Web Browser	Running and testing the web application
Web Technologies	Development of the application interface and functionality

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The FinSight AI application was implemented as a web-based system. The Portfolio Management module was developed to manage investment records through add, view, update, delete, search, and filter operations. The Financial Analysis module processes investment values and calculates total investment, current value, profit/loss, and ROI. The Risk & Recommendation module applies predefined prototype rules to generate risk classifications and basic investment insights. A dashboard provides graphical representations of portfolio information.

Hardware Implementation

No specialized hardware or physical fabrication was required for the project. A standard computer/laptop was used for application development, dataset preparation, execution, and testing.

Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Visual Studio Code	Source-code development and project organization	Development
Web Browser	Running and testing the web application	Implementation & Testing
HTML/CSS	Web interface structure and presentation	Development
JavaScript	Application functionality and calculations	Development
Investment Dataset	Portfolio data processing and testing	Dataset Preparation & Testing
Chart/Visualization Tools	Portfolio graphs and visual analysis	Implementation
Git / Version Control	Managing project source files and development changes	Development

Implementation Evidence

Figure 4.1: FinSight AI Financial Portfolio Dashboard

[INSERT YOUR MAIN DASHBOARD SCREENSHOT HERE]

4.3 Test Plan & Verification

The testing process was planned to verify whether the implemented system satisfies the defined functional and non-functional requirements. Functional testing was performed using sample investment records, while calculation verification was carried out by comparing application outputs with manually calculated expected values.

Functional Test Plan

Requirement	Test Method	Acceptance Criterion
Add investment	Enter valid investment details and submit	Investment record is successfully added
View investments	Open portfolio management screen	Stored investment records are displayed
Update investment	Modify an existing investment	Updated information is displayed correctly
Delete investment	Select an investment and delete it	Selected record is removed
Search investment	Enter an investment name	Matching investment is displayed
Filter investment	Select an investment category	Relevant category records are displayed
Calculate investment amount	Enter quantity and purchase price	Correct investment amount is calculated
Calculate current value	Enter quantity and current price	Correct current value is calculated
Calculate profit/loss	Compare investment and current values	Correct profit/loss is displayed
Calculate ROI	Apply ROI calculation	Correct ROI percentage is displayed
Risk classification	Select different investment types	Appropriate prototype risk category is displayed
Recommendation	Process portfolio information	Basic recommendation/insight is generated
Dashboard visualization	Open dashboard	Relevant portfolio information is represented graphically

Specification Verification

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Portfolio record management	Add/View/Update/Delete	Implemented	Pass
Search and filtering	Search and filter records	Implemented	Pass
Investment calculation	Correct formula	Verified using sample data	Pass
Current value calculation	Correct formula	Verified using sample data	Pass
Profit/Loss calculation	Correct result	Verified using sample data	Pass
ROI calculation	Correct percentage	Verified using sample data	Pass
Risk classification	Predefined categories	Implemented	Pass
Recommendation	Basic portfolio insight	Implemented	Pass
Dashboard visualization	Graphical output	Implemented	Pass
Web accessibility	Application runs through browser	Verified	Pass

4.4 Results

The implemented FinSight AI prototype successfully integrates portfolio management, financial analysis, risk classification, recommendation, and dashboard visualization. Functional testing confirmed that investment records can be maintained through the Portfolio Management module and that the stored information can be processed by the analysis components.

Sample Portfolio Results

Investment	Quantity	Purchase Price	Current Price	Investment Amount	Current Value	Profit/Loss	ROI
TCS	10	₹2,500	₹2,750	₹25,000	₹27,500	₹2,500	10%
SBI Bluechip	20	₹150	₹165	₹3,000	₹3,300	₹300	10%
Gold ETF	15	₹100	₹120	₹1,500	₹1,800	₹300	20%
Reliance	5	₹2,800	₹3,000	₹14,000	₹15,000	₹1,000	7.14%

4.5 Data Analysis & Engineering Judgment

The test data was analysed to verify the correctness of the financial calculations performed by the application. For each investment, the investment amount was calculated using quantity and purchase price, while the current value was calculated using quantity and current price. Profit/loss and ROI were then derived from these values.

For example, the TCS prototype record contains 10 units purchased at ₹2,500 per unit and a current value of ₹2,750 per unit. The calculated investment amount is ₹25,000 and the current value is ₹27,500, resulting in a profit of ₹2,500 and an ROI of 10%. The application output can therefore be compared with the manually calculated values to verify the calculation logic. The results indicate that structured investment data can be effectively processed to generate portfolio-level financial information. The graphical dashboard further improves interpretation by presenting portfolio information visually.

The risk classification component uses predefined rules based on investment type. This approach is appropriate for demonstrating the concept within the project scope; however, it does not capture all real-world factors affecting investment risk. Therefore, the recommendations generated by the prototype should be interpreted as basic educational insights rather than professional financial advice.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
To design a user-friendly web-based portfolio system	Web-based dashboard and portfolio interface	Fully Achieved
To develop Portfolio Management functionality	Add, view, update, delete, search, and filter operations	Fully Achieved
To gather and prepare investment data	Structured investment dataset prepared for the application	Fully Achieved
To implement financial calculations	Investment amount, current value, profit/loss, and ROI calculations	Fully Achieved
To model investment risk	Prototype risk classification based on investment type	Fully Achieved
To implement recommendations	Basic risk-based recommendation component	Fully Achieved
To develop graphical visualizations	Portfolio charts and dashboard graphs	Fully Achieved
To test the application	Functional and calculation verification	Fully Achieved
To evaluate system strengths and limitations	Results and engineering analysis	Fully Achieved

Compared with manual records and spreadsheet-based approaches, FinSight AI provides an integrated web interface with automated calculations, portfolio management, risk classification, recommendations, and graphical visualization. Compared with professional financial platforms, the proposed system is simpler and focused on demonstrating core portfolio-management and analysis functions within an academic project.

4.7 Strengths & Limitations

Strengths

- Centralized management of investment records.
- Supports multiple investment categories.
- Provides add, view, update, delete, search, and filter functionality.
- Automates investment and portfolio calculations.
- Provides profit/loss and ROI analysis.
- Includes prototype-based risk classification.
- Provides basic investment recommendations.
- Uses graphs and charts for easier portfolio interpretation.
- Provides a modular and extendable web-based architecture.
- Reduces dependence on manual financial calculations.

Limitations

- The prototype depends on the accuracy of user-provided investment data.
- Risk classification is based on predefined rules and does not represent complete real-world market risk.
- Recommendations are basic prototype insights and are not professional financial advice.
- Real-time market-data integration is not considered part of the current prototype unless separately implemented.
- The system does not execute actual buying, selling, or financial transactions.
- Advanced machine-learning-based prediction and personalized financial modelling are outside the current project scope.

4.8 Project Planning, Roles & Budget

Project Milestone Timeline

Phase	Major Activities
Phase 1	Problem identification and requirement analysis
Phase 2	Literature review and existing-system analysis
Phase 3	System design and module planning
Phase 4	Dataset gathering and preparation
Phase 5	Portfolio Management module development
Phase 6	Financial Analysis and Risk & Recommendation module integration

Phase	Major Activities
Phase 7	Dashboard visualization and interface refinement
Phase 8	Testing and verification
Phase 9	Results analysis and documentation
Phase 10	Final review and project submission

Roles and Contributions

Student	Assigned Responsibility	Actual Contribution
[TEAM MEMBER 1]	Module 1 – Portfolio Management & Dataset Preparation	Developed the Portfolio Management module and contributed to gathering, organizing, and preparing the investment dataset. Also performed testing of the contributed module.
[TEAM MEMBER 2]	Module 2 – Financial Analysis	Developed and implemented the Financial Analysis module, including total investment, current value, profit/loss, and ROI calculations.
[TEAM MEMBER 3]	Module 3 – Risk & Recommendation	Developed and implemented the Risk & Recommendation module, including risk classification and basic investment recommendatio

Project Budget

Item	Estimated Cost	Actual Cost
Computer/Laptop	Existing resource	₹0
Visual Studio Code	Free	₹0
Web Browser	Free	₹0
Dataset Preparation	Academic resource	₹0
Software Development Tools	Open-source/free tools	₹0
Internet/Data Usage	Existing resource	₹0
Total	₹0 / Existing Resources	₹0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

FinSight AI – Advanced Financial Portfolio Management and Intelligent Investment Recommendation Web Application was developed to address the difficulties associated with managing multiple investment records and analysing portfolio performance through separate or manual methods. The proposed system provides a centralized web-based platform that integrates Portfolio Management, Financial Analysis, and Risk & Recommendation functionalities.

The Portfolio Management module enables users to add, view, update, delete, search, and filter investment records. The Financial Analysis module automatically calculates investment amount, current portfolio value, profit or loss, and Return on Investment (ROI). The Risk & Recommendation module provides prototype-based risk classification and basic investment insights, while dashboard visualizations make portfolio information easier to understand.

The system was verified using sample investment data and functional test cases. The testing confirmed the operation of portfolio management functions and the correctness of the implemented financial calculations. The project therefore achieved its primary objective of developing an integrated prototype for investment portfolio management and analysis.

Overall, FinSight AI demonstrates how web-based technologies, structured investment data, automated financial calculations, rule-based analysis, and visualization can be combined to create a practical portfolio-management solution. The project provides a foundation that can be extended with more advanced financial-analysis and intelligent recommendation capabilities.

5.2 Future Work

The following improvements can be considered for future development of FinSight AI:

- **Real-Time Market Data Integration:** Integrate reliable financial APIs to automatically retrieve current market prices instead of depending on manually entered current values.
- **Advanced Machine Learning:** Develop machine-learning models for portfolio prediction, risk estimation, and personalized investment recommendations.

- **Personalized Risk Profiling:** Include factors such as investment objectives, investment duration, risk tolerance, and financial preferences for more personalized analysis.
- **Advanced Portfolio Optimization:** Implement portfolio optimization techniques to analyse diversification and suggest suitable asset allocation.
- **Secure Authentication:** Introduce stronger user authentication, authorization, and secure database management for protecting financial information.
- **Historical Performance Analysis:** Store historical investment values and provide performance trends over different time periods.
- **Mobile Application:** Develop a mobile version to provide convenient access to portfolio information.
- **Enhanced Visualization:** Add interactive charts, performance comparisons, and detailed portfolio analytics.
- **Automated Alerts:** Provide notifications for significant changes in investment values, portfolio performance, or risk levels.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned how to design and develop a structured web-based financial portfolio management system.
- Gained practical knowledge of investment data collection, organization, and dataset preparation.
- Learned how financial formulas can be implemented programmatically for automated portfolio analysis.
- Developed an understanding of rule-based risk classification and recommendation logic.
- Learned how graphical visualization can improve the interpretation of financial information.

Engineering & Professional Skills Developed

- Web application development and interface design.
- Structured dataset preparation and data management.
- Functional testing and verification of software modules.
- Problem-solving and debugging during application development.
- Technical documentation and engineering report preparation.

- Team coordination and responsibility management.
- Evaluation of design alternatives and engineering trade-offs.

Individual Reflection

Most difficult engineering problem encountered and how it was solved:

The most difficult part of my contribution was developing the Portfolio Management module while ensuring that investment records were correctly maintained and processed. The problem was addressed by organizing the investment data into structured fields and implementing the portfolio operations systematically. Testing each operation individually helped identify and correct errors before integration.

A key engineering decision made personally and the evidence behind it:

A key decision was to implement the Portfolio Management module as an organized web-based interface rather than relying on manual investment records. This decision was supported by the requirement for centralized record management, search and filtering, and integration with the other application modules.

New knowledge acquired independently:

I independently learned how to gather and organize investment-related data for use in a financial application. I also gained practical knowledge of implementing CRUD operations, handling structured investment records, and testing portfolio-management functionality.

What would be redesigned if repeated:

If the project were repeated, I would improve the dataset preparation process and design the data structure with future real-time market-data integration in mind. I would also improve validation and security features at an earlier stage of development.

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IEEE Referencing Style

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- [2] OWASP Foundation, *OWASP Application Security Verification Standard (ASVS)*, OWASP Foundation.
- [3] World Wide Web Consortium (W3C), *Web Content Accessibility Guidelines (WCAG) 2.2*, W3C Recommendation, 2023.

[4] Python Software Foundation, *Python Documentation*, Python Software Foundation.

[5] Microsoft, *Visual Studio Code Documentation*, Microsoft.

[6] The project investment dataset prepared and organized for FinSight AI portfolio-management and testing purposes.

[7] FinSight AI project source code and application resources developed as part of the capstone project.

APPENDICES

Appendix A – Detailed Calculations

A.1 Investment Amount

Investment Amount = Quantity $\times$ Purchase Price

For TCS:

Quantity = 10 units

Purchase Price = ₹2,500

Investment Amount = $10 \times ₹2,500$

Investment Amount = ₹25,000

A.2 Current Portfolio Value

Current Value = Quantity $\times$ Current Price

Current Price = ₹2,750

Current Value = $10 \times ₹2,750$

Current Value = ₹27,500

A.3 Profit/Loss

Profit/Loss = Current Value – Investment Amount

Profit/Loss = ₹27,500 – ₹25,000

Profit = ₹2,500

A.4 Return on Investment

ROI = (Profit/Loss $\div$ Investment Amount) $\times$ 100

ROI = $(₹2,500 \div ₹25,000) \times 100$

ROI = 10%

Appendix B – Engineering Drawings / Schematics

B.1 System Architecture

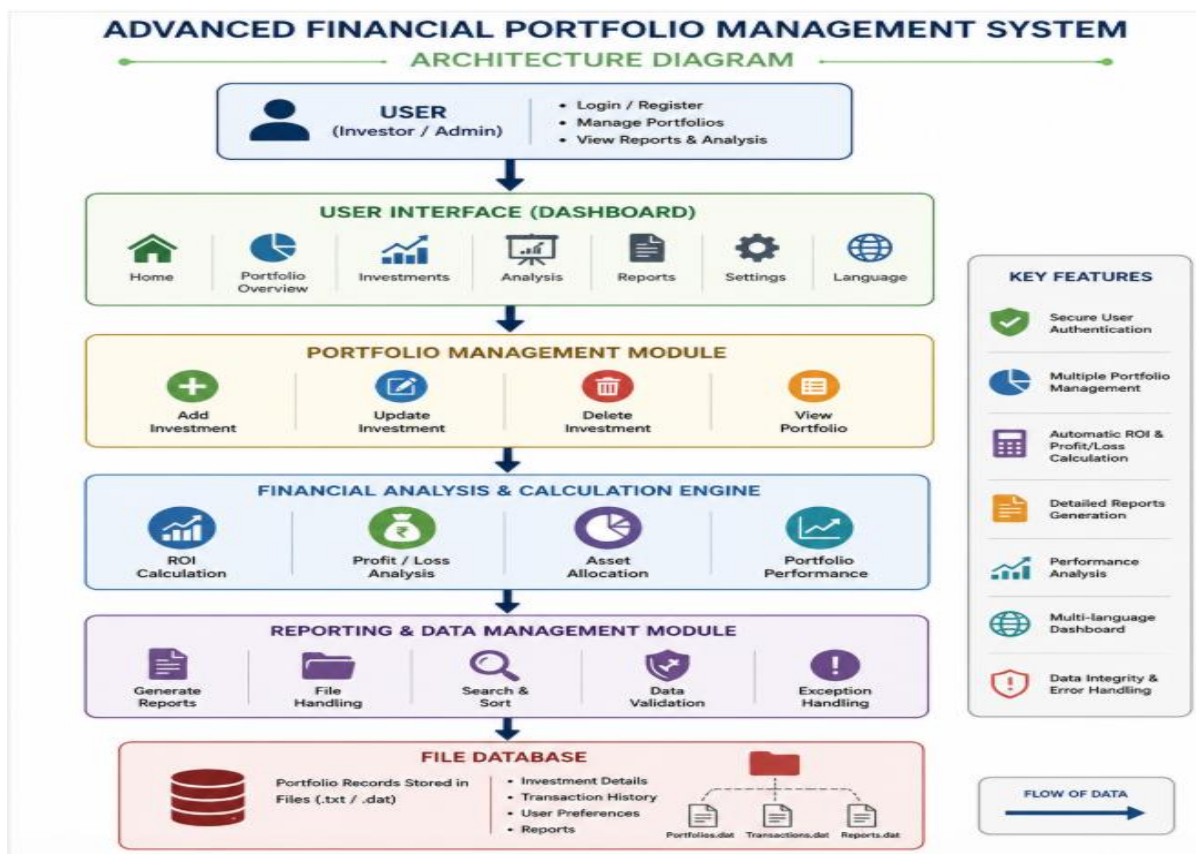


Figure B.1: FinSight AI System Architecture

B.2 Application Flow



Figure B.2: FinSight AI Application Flow

Appendix C – Source Code / Code Repository Information

The complete source code of the FinSight AI web application is maintained separately as the project implementation. The report contains only essential implementation information, while the complete project folder contains the application source files, dataset, configuration files, and supporting resources.

Project Name: FinSight AI

Application Type: Web Application

Development Environment: Visual Studio Code

Appendix D – Datasheets

No physical hardware datasheets are applicable to this software-based project.

The relevant technical information consists of the software environment, application components, investment-data fields, and tools used during implementation.

Appendix E – Experimental / Raw Data

The investment dataset used for development and testing contains structured fields such as:

Field	Description
Investment ID	Unique identifier for an investment
Investment Name	Name of the investment
Investment Type	Category of investment
Quantity/Units	Number of units held
Purchase Price	Price at which the investment was purchased
Current Price/Value	Current value used for analysis

The dataset includes investment categories such as stocks, mutual funds, ETFs, bonds, fixed deposits, cryptocurrency, and other investments.

Appendix F – Ethics / Institutional Approval

The project is an academic software prototype and does not involve human subjects, medical experimentation, or physical experimentation. No institutional ethics approval was required for the software development and testing activities.

The application is designed for educational and portfolio-management purposes. Risk classifications and recommendations generated by the prototype are not intended to constitute professional financial advice or guarantee investment returns.

Appendix G – Project Meeting / Progress Records

Phase	Activity	Outcome
1	Project topic and problem identification	FinSight AI concept finalized
2	Requirement analysis	System requirements identified
3	Literature review	Existing approaches studied
4	System design	Three-module architecture planned
5	Dataset preparation	Investment dataset organized
6	Module development	Application modules implemented
7	Integration	Modules integrated into web application
8	Testing	Functional and calculation testing completed
9	Documentation	Capstone report prepared
10	Final review	Application and report reviewed

Appendix H– Individual Contribution Evidence

Individual Contribution

StudentName: Sravanthi Duggirala

Register Number:192425315

Primary Contribution:

My primary contribution to the project was the development of **Module 1 – Portfolio Management** and the **gathering, organization, and preparation of the investment dataset** required for application development and testing.

Contribution Evidence

- Designed and implemented the Portfolio Management functionality.
- Worked on adding investment records.

- Worked on viewing investment records.
- Worked on updating investment records.
- Worked on deleting investment records.
- Implemented/search and filtering functionality.
- Gathered and organized investment-related dataset information.
- Prepared sample data for application testing.
- Performed functional testing of the contributed module.
- Contributed to technical documentation and final report preparation.

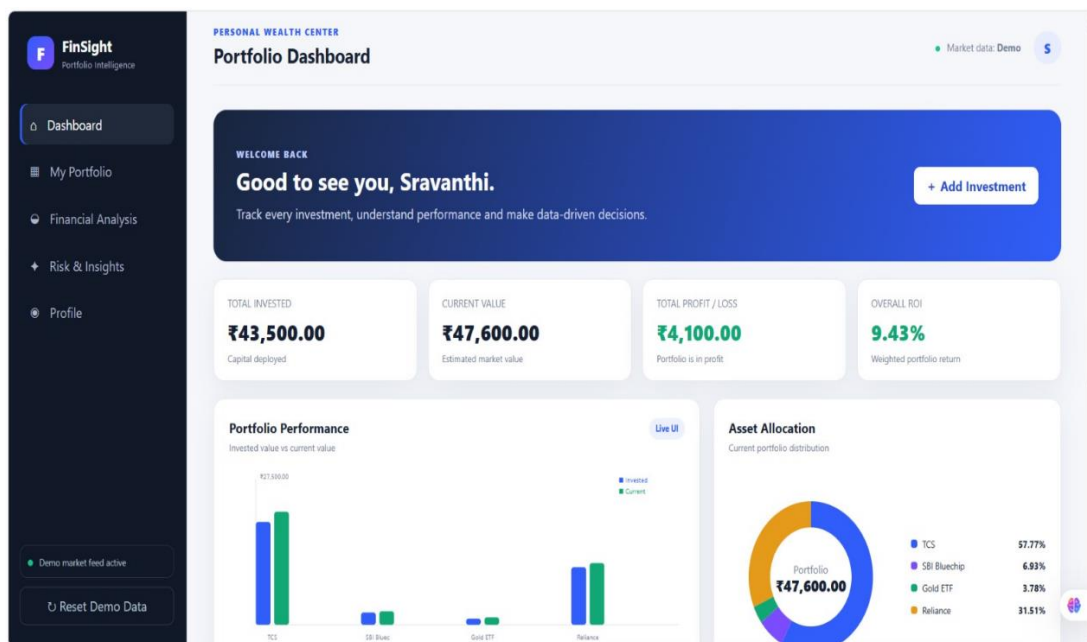


Figure J.1: Individual Contribution – Portfolio Management Module

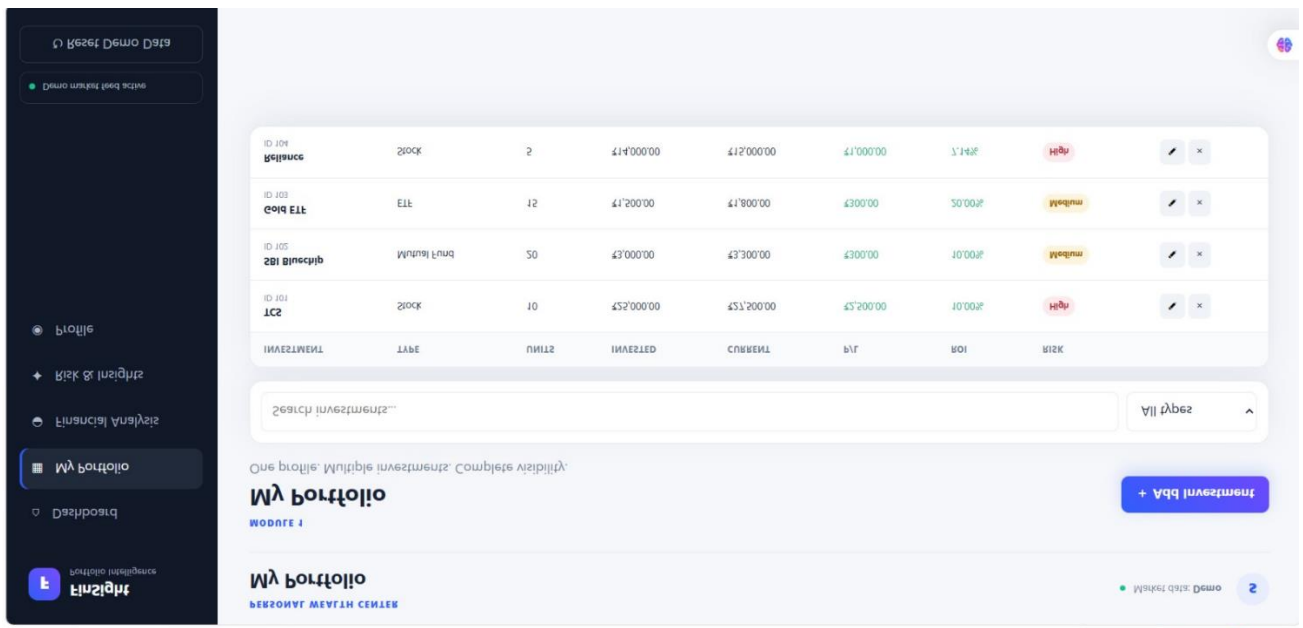


Figure J.2: FinSight AI – Module 1 Portfolio Management Interface

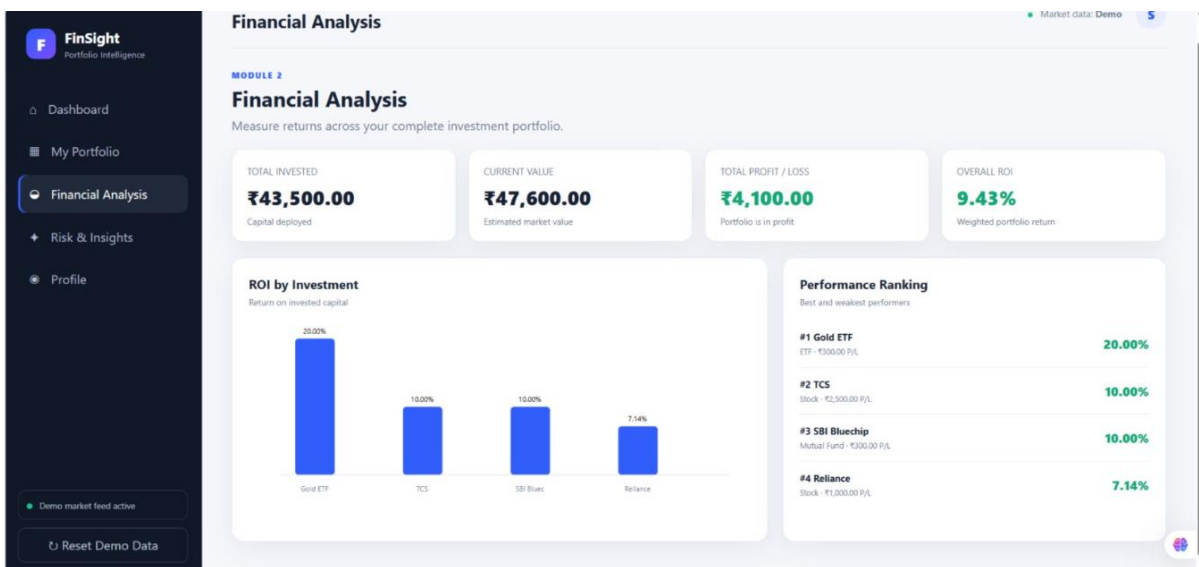


Figure J.3: FinSight AI – Financial Analysis and Portfolio Performance Dashboard

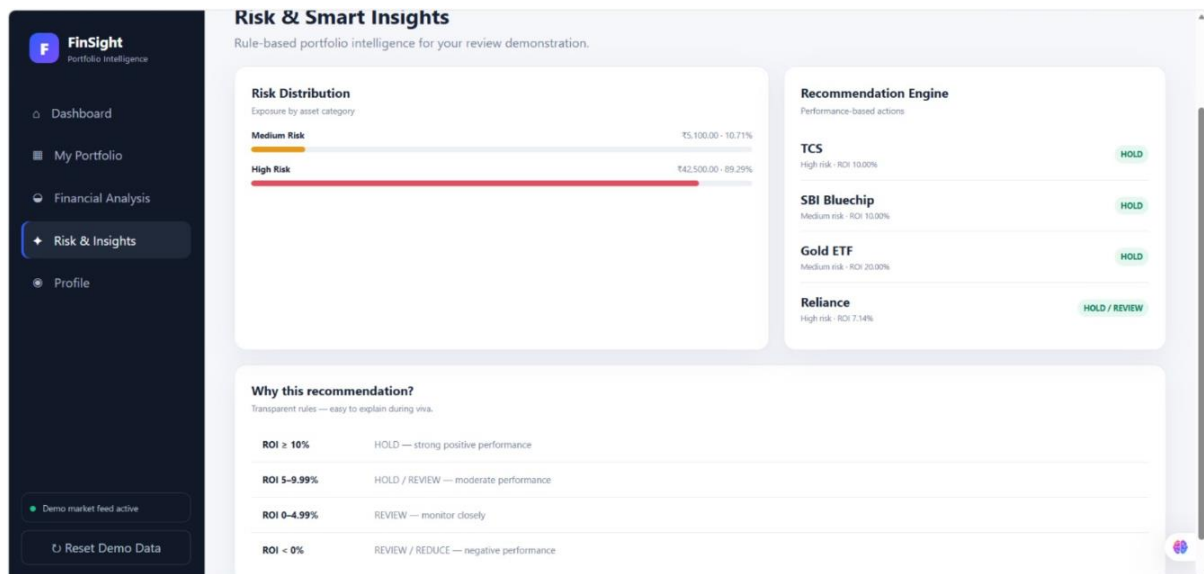


Figure J.4: FinSight AI – Risk Analysis and Smart Investment Recommendations

PERSONAL WEALTH CENTER
User Profile

Market data: Demo

ACCOUNT
User Profile
All investments belong to this single portfolio owner.

Sravanthi
sravanthi@example.com

User ID: 1001
Full Name: Sravanthi
Email: sravanthi@example.com

Save Profile

• Demo market feed active
Reset Demo Data

Figure J.5: FinSight AI – User Profile Management Interface